Ecologists have long known that the size of an animal or plant population and its growth rate depend on the availability of food, living space, and levels of predation. In the 1970s, British epidemiologist Roy Anderson and Australian scientist Robert May showed how parasites and infections from pathogens such as bacteria and viruses limited the size of a population. In wild sheep, for instance, the chief cause of death is lungworms, while most wild birds die from viral infections.

In ecology, the effects of disease have wider implications. Up to 40 percent of ocean bacteria are killed each day by viruses. This causes a “viral shunt,” because nutrients that would otherwise flow up the food chain to consumers revert to the bottom of the chain.

Human beginnings

Epidemiology has its beginnings in the work of physician John Snow, who witnessed a cholera epidemic in the Soho district of London in 1854. At the time, disease was thought to be caused by miasma—a sort of poisonous vapor in the air—that spread from the bodies of the dead and dying. Snow was not the first to question this theory, but he was especially suspicious of it in the case of cholera.

In 1854, Snow plotted every case of cholera on a map of Soho, and found that afflicted households collected their water from a pump on Broad Street (later renamed Broadwick). He shut down the pump, and the epidemic soon ended. This showed that cholera was a waterborne disease that humans contracted through contaminated food and drink. A decade later, Louis Pasteur’s “germ theory” proposed that diseases, as well as general rotting and decay, were the work of microorganisms.